Access To Space 2018
Horizontal Launch Team

AURORA

Group Design Project
Individual Report

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Abstract

A task was given to send a 100 kg payload into LEO as cheaply as possible, using a horizontal launch mechanism. This report outlines a part of the aerothermodynamic analysis required. Surface inclination codes were used to model the body through reentry, developing into body optimisation and finally subsonic and hypersonic CFD analysis, where excellent corroboration was obtained between hypersonic CFD and hypersonic surface inclination methods.

1 Introduction

1.1 Background

New developments in computational power, material science, hypersonics and propulsion are encouraging mankind to explore further into space and at a much lower cost than ever before. The aim of this project is to drive down the cost of payload insertion into a low-Earth orbit using a horizontally-launched spacecraft. This involved a multidisciplinary approach to find a feasible configuration from which to produce the first iteration of design.

1.2 Outline

Several concepts for launch configurations are initially identified and briefly discussed with a business perspective. Once an overall profile is decided on, additional selection of the baseline configuration is considered with a more technical approach: identifying key reentry parameters and making a selection based on relative importance. The most significant features of the baseline configuration are then optimised to produce the first iteration of the aerothermodynamic design. Additional analyses are carried out to assess the subsonic performance of the reentry vehicle, in order to assist other teams where launch and landing performance is required. Several methods are considered for aerodynamic analysis in both the hypersonic and subsonic regimes, each of which has limitations in terms of cost, time and suitability.

2 Baseline Conceptualisation

2.1 Launch Profile

Several launch configurations were investigated by the entire team and may be summarised as: single-stage, dual-stage and balloon. The single-stage option has a mission profile closely resembling Skylon - a conceptual spaceplane with SABRE engines, developed by Reaction Engines Limited. SABRE engines are supposedly able to operate throughout the subsonic and supersonic regimes in 'air-breathing' mode, then switching over to 'rocket-mode' above Mach 5. The engine and spaceplane are both in development and may be considered the furthest of the three from maturity. An independent business analysis conducted by the European Space Agency and London Economics [3] show that initial investment into the launch vehicle technology would be enormous and would likely require significant collaboration within European states. Furthermore, additional restrictions (6.3 tonne payload compared to 100 kg specified; and a 5.9 km runway requirement) lead to a lack of suitability.

Secondly, the balloon option was discovered to potentially have the lowest cost-per-kilogram by using an expendable helium balloon to bring the rocket to an altitude where density (and hence drag) is reduced However very little data is available on such a system and it was decided that potential growth of such an industry would not be particularly valuable. A combination of these logistical problems, combined with technical issues of balloon control made this option infeasible.

Finally, a dual-stage configuration consisting of an aircraft with a mounted orbital vehicle was considered to have the best compromise between industrial growth and technical feasibility. Such systems already exist: the Pegasus rocket by Orbital ATK is able to deliver a 400kg payload into lower Earth orbit (LEO) following a drop from a carrier aircraft at 40,000 ft [4]. It was decided that a reusable shuttle be used instead of an expendable rocket to decrease cost-per-launch. This would provide a more innovative service and may even be expanded to de-orbit defunct satellites. The rapid turnaround time of the carrier aircraft as well as the reusability of a shuttle provided a well-defined business case for the dual-stage configuration, with a suitable amount of data to draw from. Further justification of overall lanuch configuration may be found in [5].

2.2 Shuttle Concepts

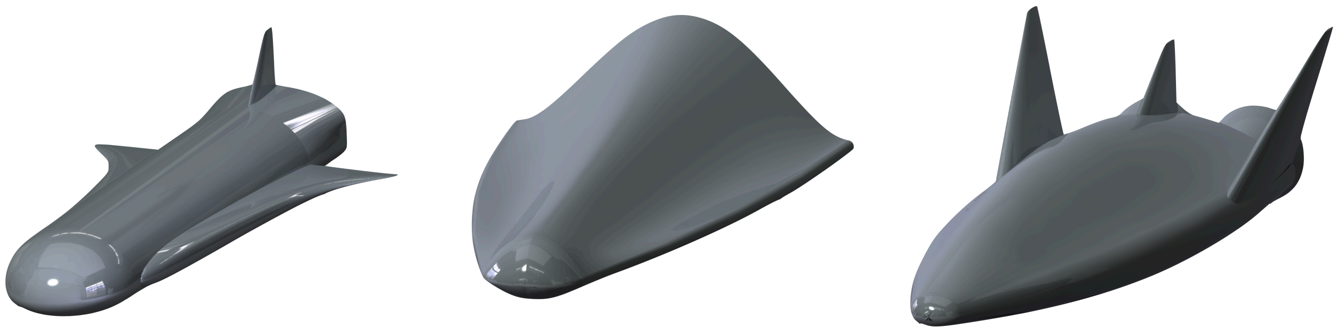
Following a review of existing concepts, three broad categories of shuttle configuration were found: delta wing, hypersonic gliding vehicle (HGV) and blended wing-body (BWB). For each case, examples were found and used to size rough dimensions in order to begin an initial comparison.

2.2.1 Delta Wing

The delta wing is already well-known due to the space shuttle orbiter, however a delta wing configuration has also been used successfully by another spacecraft. The X-37B has successfully been launched and returned [6], and is of a similar scale to that required by the project. Initial sizing of the delta-wing configuration was used with limited data from the X-37B.

2.2.2 Hypersonic Gilding Vehicle

The HGV is designed for gliding at high altitudes at hypersonic speeds. For reentry, an extended glide has the effect of reducing peak heat flux by spreading energy dissipation over a greater period of time. The design of the HGV was simply adapted from [7]. Additional geometries for wings are specified in [8] and were also analysed. The inherent design of the HGV means that it is not concerned with reusability. The design was therefore adapted by rounding the nose from the original design in order to lower heat flux at the nose.



(a) Delta wing (b) Hypersonic Gliding Vehicle (c) Blended wing-body

Figure 1: Potential shuttle configurations

2.2.3 Blended Wing-Body

A BWB combines the effect of a lifting body along with the added lift of wings at high dihedral. The wings would provide enough lift at low speeds for launch and landing, whilst the heavily-reinforced fuselage would withstand the loading during reentry. The BWB design is based of Sierra Nevada's Dreamchaser, which in turn is based on the HL-20 lifting body design[9]. The BWB is therefore also considered a lift-body: the fuselage generates lift during subsonic flight, decreasing the wing size required.

3 Shuttle Configuration

3.1 Modified Newtonian Panel Code

The kinematic equations of a reentering body found in [10] may be reduced to a 2D problem with three states [11]. These equations may be written in terms of two parameters: ballistic coefficient (BC) and lift-to-drag ratio (L/D) and are shown in Equation 5.

$$BC = \frac{m}{C_D S} \tag{1}$$

and

$$L/D = \frac{C_L}{C_D} \tag{2}$$

Where m is the body mass, S is the wetted area, C_D is drag coefficient and C_L is lift coefficient. For reentry, wetted area is chosen over the wing area as it is the area facing the oncoming high-speed flow that produces lift (whether from the fuselage or wings). Therefore, in order to analyse the reentry characteristics of each shuttle, it was necessary to generate data on these parameters for each concept, which heavily depend on the respective geometry.

3.1.1 Theory

Reentry is dominated by the hypersonic regime and so it was necessary to find C_L and C_D using hypersonic methods. At high Mach numbers, shock waves are close to the surface of the body: the small region between the shock and the body is known as the thin shock layer. Due to this, the flow is essentially deflected by the same angle as the local inclination of the body itself. As a result, the flow may be considered to lose all of its momentum perpendicular to the local surface, whilst retaining momentum parallel to the surface. After considering the conservation of momentum and mass, the pressure coefficient, C_P, may be written as a function of the local surface inclination, θ, and the pressure coefficient behind a normal shock $C_{P_{max}}$ at the freestream Mach number [12]:

$$C_P = C_{P_{max}} \sin^2 \theta \tag{3}$$

This is known as modified Newtonian theory: a surface inclination method, due to dependence on θ and not on the surrounding flow. When considering modified Newtonian theory, the fluid particles only impinge on the 'visible area' (windward side) of the body, i.e. any area exposed to the incoming flow. Since areas in the aerodynamic shadow (leeward side) are not exposed, the fluid does not impact them and so the pressure is assumed to be equal to freestream pressure $\implies C_P = 0$ [13].

Other surface inclination methods may also be used, such as tangent wedge and shock-expansion theory. However, these methods are only applicable to slender bodies, whereas modified Newtonian theory is suitable for bluff bodies, which is typical for a reentry design. Later in the design process, the tangent wedge method was used to analyse the slender wings of the selected shuttle, whilst modified Newtonian was kept for the fuselage. This is discussed in [14].

3.1.2 Application

Local surface inclination methods evidently depend on the angle between the freestream and the body surface. It was therefore necessary to discretise the whole surface into a series of surface panels. For a three dimensional body, the angle of each panel must be found by finding the three-dimensional vector dot product between the surface normal vector, $\boldsymbol{n}$ and the freestream velocity vector, $\boldsymbol{v}_\infty$:

$$\sin\theta = \frac{\boldsymbol{v}_\infty}{|v_\infty|} \cdot \boldsymbol{n} \tag{4}$$

The surface of the shuttle body was discretised from a CAD model by producing a point cloud output. Using the MATLAB function `alphaShape`, the surface was meshed into triangular panels, each with a normal vector in fixed coordinates. It was necessary to split certain parts into separate surfaces to prevent meshing between those in close proximity. Discretisation of each surface along with corresponding pressure coefficient distribution may be found in Appendix A.

It may be noted that the testing conditions are at a Mach number of 20 and an angle of attack of 45°. This constant angle was considered to be typical during the hypersonic regime of reentry [15]. Afterward, during reentry optimisation, it was decided in [14] to use an angle of 45° anyway. The Mach independency principle of hypersonic flow may be used to evaluate C_L and C_D as constant at hypersonic speeds [12]. Therefore, only a single set of conditions are needed to evaluate L/D and BC for each shuttle.

3.2 Reentry

As mentioned before, two-dimensional modelling of an uncontrolled reentry was carried out by considering three states: altitude h, velocity v and flight path angle γ, shown below in Figure 2. The governing ordinary differential equations shown below are adapted from [10], where ρ is density (which is heavily dependent on h) and R is the equatorial radius of the Earth. BC and L/D are inputs from the modified Netwonian method.

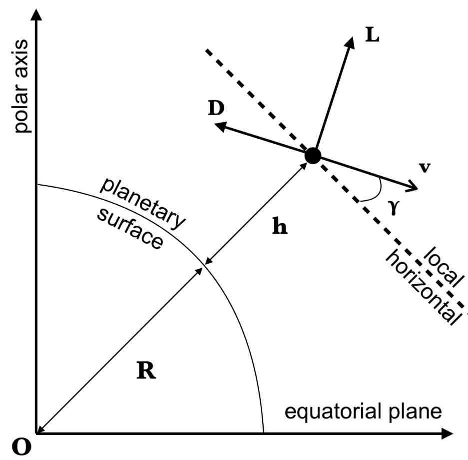


Figure 2: Coordinate system of the reentry parameters in 2D. Diagram simplified from [1]

$$\begin{aligned}
\frac{\mathrm{d}h}{\mathrm{d}t} &= v \sin\gamma \\
\frac{\mathrm{d}v}{\mathrm{d}t} &= -\frac{1}{2}\rho v^2 \frac{1}{BC} - g\sin\gamma \\
\frac{\mathrm{d}\gamma}{\mathrm{d}t} &= \frac{1}{2}\rho v^2 \frac{1}{BC} L/D - (\frac{g}{v} - \frac{v}{R+h})\cos\gamma
\end{aligned} \tag{5}$$

A MATLAB lookup table of the 1976 U.S. standard atmosphere was used to model the variation of atmospheric properties with altitude. This modelling is further dicussed in [16]. Using MATLAB's `ode45`, the solution was time-marched through the reentry, recording the each state at small timesteps. Initial conditions were provided by the control team: $h(0)$ = 150 km and $v(0)$ = 7112 ms^{-1}.

3.2.1 Skipping

Due to the absence of control inputs, the trajectory followed is simply as a result of aerodynamic forces acting on the body. As the shuttle descends, atmospheric density and hence lift increases. Eventually, enough lift is produced to cause the shuttle to gain altitude. Such an ascent is not sustainable and the shuttle begins to fall again in what is essentially a large-scale phugoid called *skipping*.

During reentry, the space shuttle orbiter performed S-turn manoeuvres to control this effect [15]. By changing bank angle of the spacecraft, the lift vector is turned through the same angle, decreasing the vertical component of lift, reducing the skipping effect.

However, skipping should not regarded as an undesirable phenomenon. A skipping reentry was used during the Apollo missions to decrease heating of the reentry capsule by extending range. An optimised reentry trajectory for the final shuttle was developed in [17] later in the project. However, uncontrolled skipping led to large velocities and therefore very large heat flux and dynamic pressure results. For the purposes of selection, skipping was eliminated entirely by holding bank angle constant. It was found that a constant bank of angle of around 75° was required for each shuttle concept to perform a reentry without skipping.

3.3 Selection

With velocity and altitude data for each of the shuttles, parameters such as dynamic pressure, P_d and heat flux $\dot{q}$ may be found. The heat flux equations below are simple models for convective and radiative heating ($\dot{q}_{conv}$ and $\dot{q}_{rad}$ respectively), used for selection purposes. They are simply functions of nose radius, r_n, v and ρ. A more extensive heating analysis of the final shuttle may be found in [18].

$$P_d = \frac{1}{2}\rho v^2 \tag{6}$$

$$\dot{q}_{conv} = k\left(\frac{\rho}{r_n}\right)^{\frac{1}{2}} v^3 \tag{7}$$

$$\dot{q}_{rad} = c{r_n}^a \rho^{1.22} f_E(v) \quad (8)$$

Where $\dot{q}_{conv}$ is convective heat flux from [19] and $\dot{q}_{rad}$ is radiative heat flux from [20]. Constants are defined as: $k = 1.762 \times 10^{-4}$, $c = 4.736 \times 10^4$, $a = 1.07 \times 10^6 v^{-1.88} \rho^{-0.325}$ and $f_E(v)$ may be taken as a maximum of 1.5 (from [20]). It was found that radiative heating is essentially negligible throughout reentry, and the total heat flux may be approximated as the convective contribution only.

3.3.1 Quantitative Selection

Consideration of heat load (due to convective heat flux) and mechanical load (due to dynamic pressure) must be considered when selecting the shuttle. The reentry loading histories are shown in Figure 3. In conjunction with Table 1, it can be seen that $\dot{q}_{max}$ and $P_{d_{max}}$ decrease with BC. The maxima also occur at lower altitudes for high BC, indicating that drag is too low at high altitudes to slow the body down.

Structural mass is often one of the most significant mass fractions of reentry vehicles. Defining the non-dimensional volumetric efficiency η_v as the total usable volume divided by a power of the wetted area allows for a decrease of the structural mass. Finally, the final trajectory still involves a significant gliding phase, and so L/D has significance. Table 1 and Figure 3 show the main results of the reentry study. It is clear that the Winged-HGV and BWB are better choices for the shuttle design. They both have lower values of $P_{d,max}$ and $\dot{q}_{max}$ whilst maintaining a high volumetric efficiency. The conventional delta-wing performs poorly in all metrics.

3.3.2 Qualitative Selection

After discussion with the structures team, concerns were raise about the feasibility of manufacture of the HGV. The highly curved wings of the winged design would have to be replaced with straightened versions and would likely need to be drawn in to reduce the cantilever moment. Additionally, the curved nature of the fuselage (although well-defined in [7]) would be similarly difficult to accomodate. In contrast, the delta-wing configuration was considered to be the simplest design for manufacture and sizing. The blended surface of the BWB made it difficult to parameterise the upper surface, however was not as complex as the HGV.

The origin of the design must also be considered during selection. As discussed before, alterations were made to the original design as the HGV was not concerned with reusability. This is because papers used for initial sizing [7], [8] were originally in the context of a hypersonic missiles and trajectory planning to minimise the chance of interception. It may be possible that designs of the shuttle or even the shuttle itself (should the shuttle be ditched during separation from the carrier, or during the last stages of reentry) may be obtained by more belligerent third-parties.

Highly lifting (asymmetric) reentry bodies are far less common than simpler axisymmetric shapes. It would be likely that the entire team would need to draw upon historical data for initial sizing and order of magnitude checks. During the initial research on concepts, it was clear that the only design with extensive published data is the space shuttle orbiter, meaning the delta-wing was again advantageous in this respect. In contrast, the HGV is purely conceptual and the military nature means that any testing remains highly confidential. The BWB takes its design from a number of experimental aircraft (namely the X-24A and the more modern Dreamchaser) as well as conceptual spacecraft such as the HL-20. Data for these vehicles is not as well documented as the space shuttle orbiter, but is still significant.

With all the above selection criteria in mind, the blended wing-body design was selected as the best compromise between performance and design practicality.

Table 1: Quantative parameters found for each shuttle reentry

Shuttle Concept	BC (kg/m^2)	L/D	C_L	C_D	$P_{dyn,max}$ (kPa)	$\dot{q}_{max}$ (W / cm^2 /s)	η_v
Delta-wing	87.26	0.895	0.428	0.478	5.37	146.8	0.0293
HGV	48.79	0.865	0.544	0.629	3.09	110.8	0.0421
Winged-HGV	39.45	0.893	0.508	0.569	2.48	100.0	0.0327
BWB	45.62	0.854	0.576	0.675	2.93	107.8	0.0434

Table 2: Qualitative parameters taken into consideration, graded on a low / medium / high scale

Shuttle Concept	Manufacture Difficulty	Ethical Concerns	Historical Data	Simulation data
Delta-wing	Low	Low	High	High
HGV	High	High	Low	Low
BWB	Medium	Low	Low	Medium

4 Shape Optimisation

Once an initial geometry had been decided upon, it had to be optimised with respect to the constraining variables considered before (such as $\dot{q}_{max}$). The shuttle was split into three main parts: fuselage, wings and nose. The fuselage shape was found to be limited by the volume required by the structure and internal systems, with little room to adjust. On the other hand, the nose geometry was found to be dominated by heating effects and was analysed with this in mind.

4.1 Nose

The nose was considered as two half-ellipsoids on top of each other (known as an ellipsled) which share a cross-sectional semi-major axis, while the ratios of semi-minor axes were varied. Several constraints were imposed upon the the nose geometry: the overall height and width must match that of the fuselage (1.7 m and 2 m respectively, as determined by the fuslage internals) and the internal volume of the nose must accommodate that of the systems and nose landing gear. An initial sizing of 2m^3 was given, but a later systems evaluation suggested this may be reduced in the future. The nose shape is generalised in Figure 4.

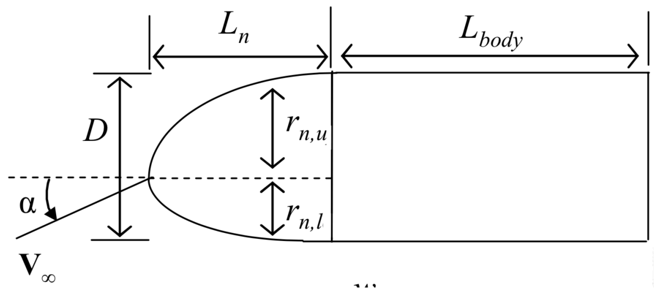
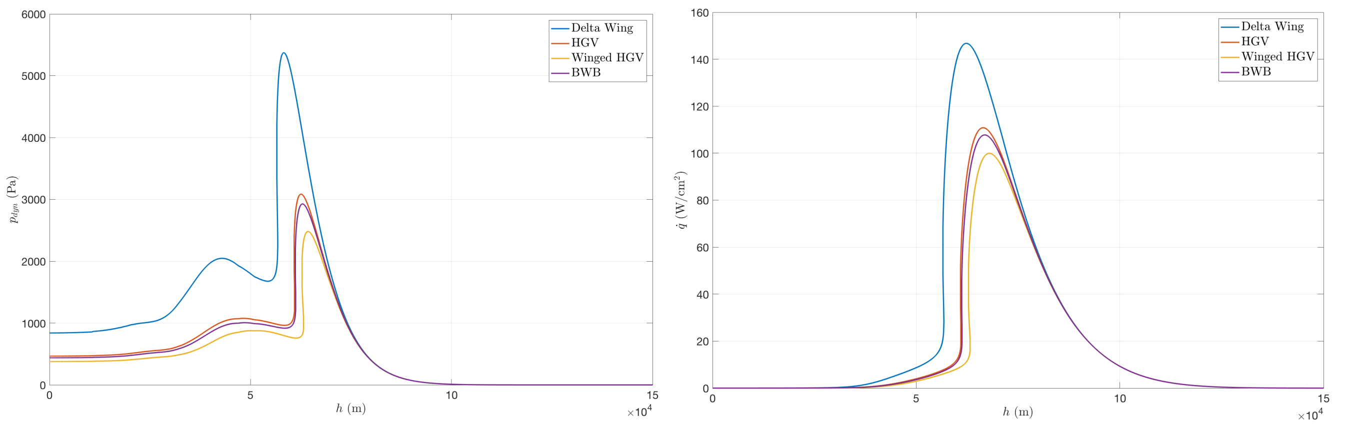


Figure 4: Parameteriseation of the nose geometry [2]

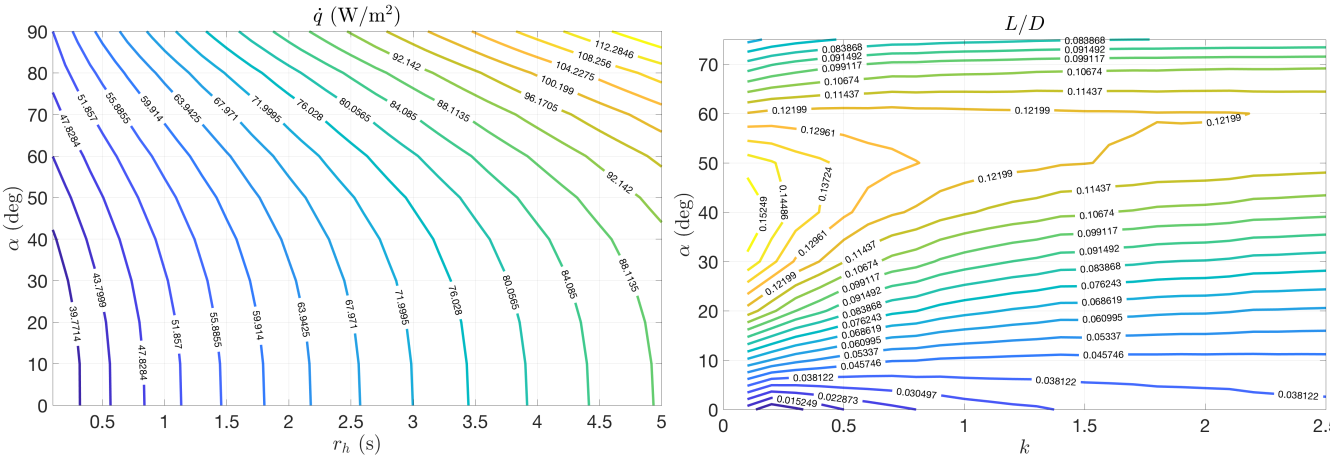
Firstly, the volumetric constraint was considered. The shape was formed by plotting a discretised form of the ellipsled for various radius ratios, $k = \frac{r_u}{r_l}$ and and nose lengths, L_n. The discretised surface was created using the `alphaShape` function. It was found that the volume of the



(a) Dynamic pressure reentry history

(b) Stagnation point flux reentry history

Figure 3: Dynamic pressure and heat flux for each shuttle concept.



(a) Lines of constant $\dot{q}$ for varying α and k

(b) Lines of constant L/D for varying α and k

Figure 5: Contour plots of L/D and $\dot{q}$ for a grid of α and k values.

nose cone is effectively independent of the height ratio for a given length but scales linearly with length, shown in Appendix B. This implies that the volumetric constraint requires a nose length of approximately 2 m.

The remaining parameter to find was the height ratio, k. The nose is a relatively large region of high pressure during reentry; changing the nose geometry was thought to also change the reentry characteristics. Since reentry trajectory was still unknown, the same trajectory approach was used as before: an uncontrolled descent at constant bank angle was time-marched for every geometry.

Each nose shape was combined with the fuselage surface, where BC and L/D were evaluated using the modified Newtonian code developed earlier. Height ratios in the range of 0.1 to 5 were considered were considered for a range of angles of attack α, to see if an optimum combination may minimise stagnation point heat flux. This was evaluated again using Equation 7, the Sutton-Graves stagnation point heating model. Here, the nose radius r_n is taken to be the lower height, r_l. This was considered to be a good enough indicator until a more sophisticated heating model was developed. Additionally, η_v, L/D and BC were investigated. The contour values in Figure 5 were not and should not be taken at face value, due to the simplicity of both the trajectory and heat flux models used, but more of an indicator during early design. It was expected that the low curvature corresponding to low height ratio would lead to decreased stagnation point heat flux and this is corroborated by the contour plot. L/D also shows a very clear maximum at a low height ratio near 40deg angle in Figure 5b. η_v and BC showed little dependence on height ratio at higher angles of attack and were not included. From the results, it can be seen that a nose height of less than one is desired. Following a significantly more comprehensive heating analysis in [18], it was agreed that a low value of k gave better heating performance. A height ratio of 0.5 was selected as a smaller value would cause the shape to tend toward a sharp corner on the upper surface which is also not desired.

5 Subsonic CFD

5.1 Motivations

Due to the reliance on lift produced by the shuttle during landing (and launch to a lesser extent), multiple teams required information on the aerodynamic characteristics of the shuttle at low speeds. Since the design and optimisation of the spacecraft considered hypersonic conditions as the constraining case, the shuttle was later analysed for subsonic performance. Future iterations may use the subsonic characteristics to tailor the design if required.

As shown in [14], the surface inclination methods lose accuracy at $M < 3.5$. Therefore another approach is required for subsonic / low supersonic flowfields. Many possible methods were considered throughout the project. Wind tunnel testing was strongly considered in both supersonic and subsonic tunnels. However a lack of similarity rules for the supersonic tunnel [18] and a lack of time for the subsonic tunnel [14] meant that these plans did not materialise. Subsonic panel codes were also considered. Inviscid panel codes (such as the Tornado Vortex Lattice method) were deemed unsuitable for this application. This is because such codes are not able to accurately determine separation or interaction between flows (such as the wake of the wing with the flow over the fuselage). Further discussion on panel codes may be also be found in [14]. It became clear that in order to produce data, the CFD software STAR-CCM+ would be required. It was highly desired to validate the CFD results with other methods, however for the same reasons as above, this could not happen. CFD Although several sets of data were requested by different groups, time limitations meant that only landing data for the control and groundlaunch teams could be provided. Additional data for the launch configuration with the external fuel tanks should be found in the next design iteration.

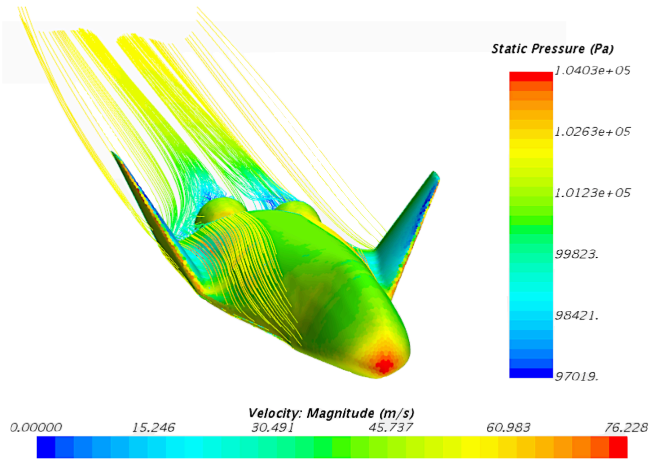


Figure 6: Pressure distribution on the surface of the shuttle coming in to land. Note the streamlines accelerate over the body, and the recirculation region at the back of the nozzles.

5.2 Landing

One of the goals of the project was to land on a commercial runway in order to make use of existing facilities and bring down costs. It was decided by the groundlaunch team to consider landing at 60 ms^{-1}. A maximum angle of 16° was specified, to avoid the nozzle striking the runway. To find a suitable angle of attack α at which this may occur, CFD was run at 60 ms^{-1} for 4 angles (-5° to 10°). From this, C_L and C_D curves were produced, shown in Figure 7. Since the body has a very low aspect ratio, the flow is certainly 3D. To reduce meshing and computation time, a half-body was used along with a symmetry plane. Due to time constraints, steady simulations only were run.

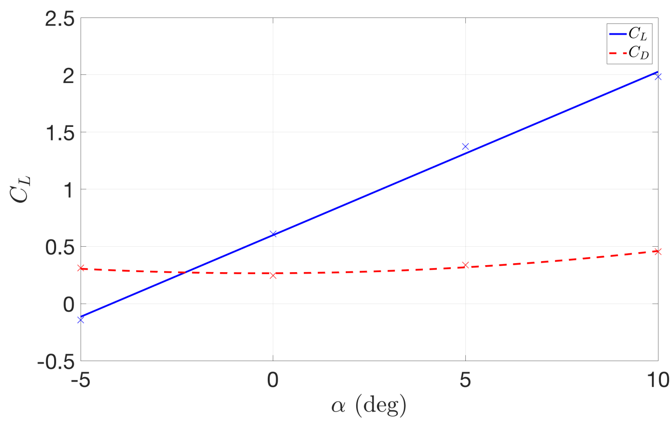


Figure 7: C_L and C_D at 60ms^{-1} and varying angles of attack.

To support a landing weight of 12kN, a C_L of of 0.78 is required, indicating $\alpha < 5°$. Figure 7 and the implication for α is obviously not very realistic. This may be because C_L and C_D are functions of S, the reference area, taken to be the wing area of 7 m^2. However a blended wing-body produces significant lift on the fuselage as well; inclusion of the the fuselage lifting area would perhaps produce a more accurate range of C_L and C_D. Although it is not certain that the flow solution is valid, examining Figure 8 indicates that the flowfield is at least reasonable. Flow is accelerated over the upper surface of the shuttle and separates over the back side, leaving a recirculation bubble. Even so, there are more robust ways to ensure the solution of the simulation is accurate.

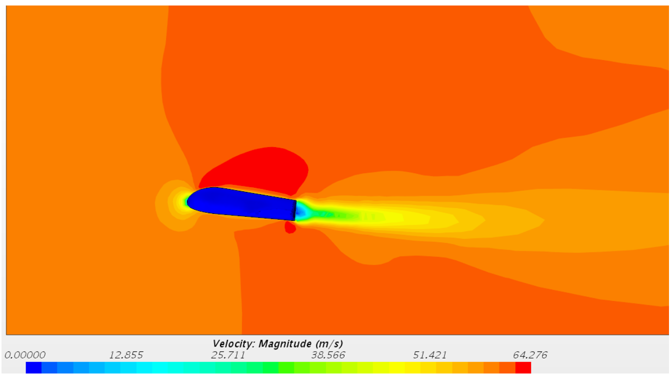


Figure 8: Veloctiy profile around the centreline of the shuttle at 60ms^{-1}

5.3 Accuracy

5.3.1 Numerical Accuracy

As discussed before, validation of the results could not take place by comparison with other methods. However, in order to ensure the flowfield was captured as accurately as possible, the mesh and numerical solver were considered carefully. CFD runs by iteratively solving transport equations across a discretised domain. Discretisation may also occur in time is the solution is unsteady. In order to acquire an accurate solution, the discretisation, or mesh, must be fine enough to capture all aspects of the flow. However, decreasing the mesh size (and hence increasing the number of cells in the domain) comes at a computational cost. To arrive at the correct solution in an efficient manner, the mesh must be refined in regions where it is necessary, such as areas of high curvature and high gradients.

Mesh refinement around the body and in the wake was achieved using volumetric controls. An example of this is shown in Figure 9, where volumetric refinement is discussed later. A polyhedral mesh was used throughout the domain. At low speeds such as this, skin friction drag becomes significant and hence the boundary layer must captured appropriately. This is accomplished using a fine prism layer mesh. In order to size the prism layer, the appropriate y^+ (non-dimensional wall distance) value was required. From [21], it was found that a reasonable y^+ range is 30-300 at low speeds. By taking a nominal value of y^+ = 50, the thickness of the first prism layer may be found. Rearranging the definition of y^+:

$$y = \frac{y^+\mu}{u_\tau \rho} \tag{9}$$

Where u_τ is the wall friction velocity, as described in [22]. u_τ may be found by a series of back substitutions:

$$\begin{aligned} u_\tau &= \sqrt{\frac{\tau_w}{\rho}} \\ \tau_w &= \frac{1}{2} c_f \rho U_\infty^2 \\ C_f &= \frac{0.664}{\sqrt{Re_l}} \\ Re_l &= \frac{\rho U_\infty L}{\mu} \end{aligned}$$

Where τ_w is the shear stress on the shuttle surface, c_f is the coefficient of friction and Re_l is the length-based Reynolds number. The expression for c_f is given by the Blasius solution for a laminar boundary layer along a flat plate in zero pressure gradient. The fuselage is relatively flat, however the pressure gradient along the fuselage is likely to be adverse. This would occur if the pressure distribution over the fuselage mimicks that of a wing and the suction peak is near the front. The Blassius solution was therefore only used as a first approximation, and subject to revision. Since the flow is dominated by factors along the fuselage, the length scale L was taken to be the fuselage length of 7 m when calculating Re_l. Therefore:

$$\begin{aligned} Re_l &= 28.6 \times 10^6 \\ \implies c_f &= 1.24 \times 10^{-4} \\ \implies \tau_w &= 0.274 \\ \implies u_\tau &= 0.4728 \\ \implies y &= 1.6\text{mm} \end{aligned}$$

Checking order of magnitude of τ_w by assuming a linear boundary layer profile and using the definition of wall shear stress:

$$\tau_w = \mu \frac{\partial U}{\partial y} \approx \mu \frac{\Delta U}{\Delta y}$$

Using the value of τ_w above gives a boundary layer thickness (Δy) of 4.0 mm, which seems within an appropriate order of magnitude for the flow. Following the results of the analysis, the first prism layer was placed at y = 1.6 mm, with 6 additional prism layers to ensure the boundary layer was represented well.

In addition to the prism layer refinement, an appropriate turbulence model must be selected. A realisable $k - \epsilon$ turbulence model is well-studied and is appropriate for transitional flows and vortex shedding [23], both of which is expected for the shuttle landing. The turbulence intensity I and turbulent length scale L_t were calculated from Equations 10 and 11 as described in [21].

$$I = \sqrt{\frac{1.5 v_t^2}{U^2}} \tag{10}$$

$$L_t = 0.07 D_h \tag{11}$$

Where v_t is the turbulent velocity which is approximated as 10% of the freestream velocity, and D_h is the hydraulic diameter, both defined in [21].

5.3.2 Mesh Accuracy

To confirm that the mesh quality did not affect the accuracy of the results, a mesh independence study was carried out. The conditions were kept the same: 60 ms^{-1} at 5° angle of attack. The overall mesh structure was also kept the same, with volumetric controls refining the mesh near the body. The controls were scaled with the base size. Nine meshes were tested: a base size of 0.4 - 2 m at 0.2 m intervals. Additional meshes were attempted at 0.2 m and 0.1 m base size, however the desktop computers did not have sufficient computational power. As mentioned before, the shuttle is symmetric in the x-y plane. Imposing a symmetry condition halves the computational domain, allowing for greater refinement. Example sections of each mesh is shown in Figure 18 in Appendix C.

From Figure 9, it can be seen that although the mesh of 9b is 5 times finer than that of 9a, the velocity field is relatively consistent (albeit more rough). Both velocity fields indicate acceleration over the upper surface of the shuttle, low speed near the nose and a wake behind the body with a very low-speed recirculation region. These observations seem sensible and demonstrate a degree of numerical accuracy of the simulations.

For each mesh, C_L and C_D were evaluated, shown in Figure 10. C_L and C_D may be seen to begin to tail off to around 1.4 and 0.4 respectively. However results of the mesh independence study were not as conclusive as hoped for. In order to demonstrate a higher degree of mesh independence, the volumetric refinement must be restricted to smaller regions and with greater care to limit the number of cells produced while still being able to reduce the base size of the mesh -since the current mesh layout could not be further refined beyond a base size of 0.4 m.

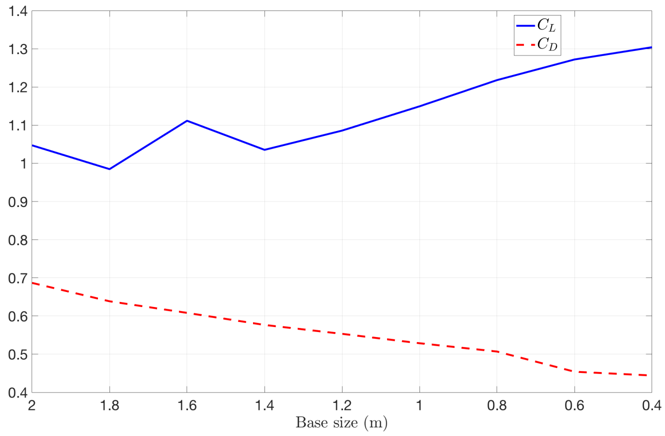


Figure 10: C_L and C_D with increasing mesh refinement

6 Hypersonic CFD

6.1 Motivations

To help evaluate the accuracy of the surface inclination code, the reentry conditions were set up in CFD as well. Following a finalised reentry trajectory from [17], the freestream conditions at four Mach numbers were found (shown in Table 3) and set up in both the STAR-CCM+ and the surface inclination code with the aim to visually compare pressure distribution and numerically compare C_L and C_D.

6.2 Methodology

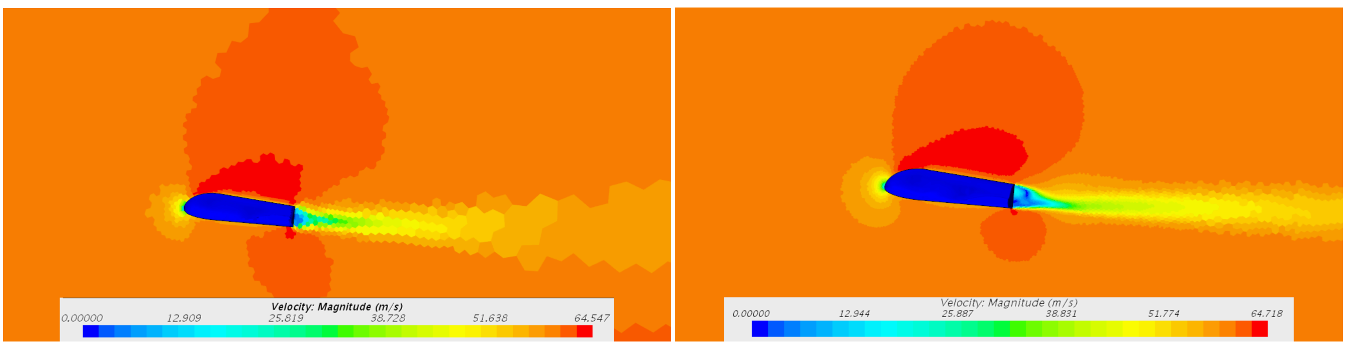
6.2.1 Computational Limitations

An axisymmetric body may be analysed in 2D and rotated about the axis of symmetry relatively easily. This cannot be done with this shuttle however, presenting a significant computational obstacle. Initially, a 2D centreline analysis at Mach 3 conditions was carried out. The mesh was small enough to easily refine and accommodate the sharp pressure and velocity gradients of the shocks. But in order to produce meaningful data, a 3D analysis is required due to the low aspect ratio and asymmetry of the shuttle. When extruding to 3D, mesh refinement was severely limited by the memory (RAM) of the desktop computers. The memory demand by STAR-CCM+ is high when: a) generating the volume mesh and b) when running the simulation. Both processes are heavily dependent on the number of cells within the mesh (as is the accuracy of the simulation). It was found that running the simulation required more memory (and is therefore more limiting) than meshing. With thanks to C. Cantwell, access was provided to run STAR-CCM+ on two remote linux clusters of the Imperial aeronautics department- `spitfire` and `hurricane`. This decreased computational time significantly as well as reducing the memory restriction down to volume mesh generation only.

6.2.2 Mesh Refinement

In order to make use of the finite memory, selective mesh refinement was again assessed. This time, the mesh was refined heavily at and behind the shock. The very large pressure and velocity gradients behind a shock mean that a mesh may not be able to capture the gradients sufficiently, leading to numerical error and/or divergence of the solution. The region where the shock is normal to the flow has a significantly greater pressure gradient than further away (i.e. the shock strength is higher), meaning that further refinement is required in this region (which coincides with the nose of the shuttle).

Now that the physical basis for regions of interest have been established, locating such regions are the next problem. For attached shocks on slender bodies, the shock wave originates from the body and moves outward at a relatively constant angle. This provides a good starting point for mesh refinement. However the shuttle is a bluff body with a detached bow shock. For a sphere-cone with



(a) Base size 2 m

(b) Base size 0.4 m

Figure 9: The velocity flowfield for the 2 m (l) and 0.4 m(r) base size

Table 3: Freestream conditions to be tested

Mach number	Altitude (km)	Velocity (km/s)	Density $\times 10^{-4}$ (kg/m^3)	Temperature (K)	Static Pressure (Pa)
3	5.61	0.96	4.71	256	34.6
5	6.36	1.54	1.81	235	12.3
7	6.77	2.11	1.02	224	6.6
10	7.32	2.91	0.46	210	2.8

a detached shock, the shock stand-off distance Δ may be found using relations found in [24], shown in Equation 12. The lower surface of the shuttle is rounded at the nose and flat behind it and so can be reasonably simplified to such a geometry. Where the nose R_n is simplified to the lower nose radius, at an angle of 45°, and the cone angle θ_{sc} is simplified to the 45° angle of attack achieved during reentry.

$$\Delta = 0.85(\frac{\rho_\infty}{\rho_{02}})R_n \tag{12}$$

Where $\frac{\rho_\infty}{\rho_{02}}$ is the density ratio of stagnation density behind the shock to freestream density, broken down using the ideal gas law and the fact that total temperature T_0 is conserved across a shock wave.

$$\frac{\rho_\infty}{\rho_{02}} = \frac{P_\infty}{P_{02}}\frac{T_{0\infty}}{T_\infty}\frac{T_\infty}{T_2}$$

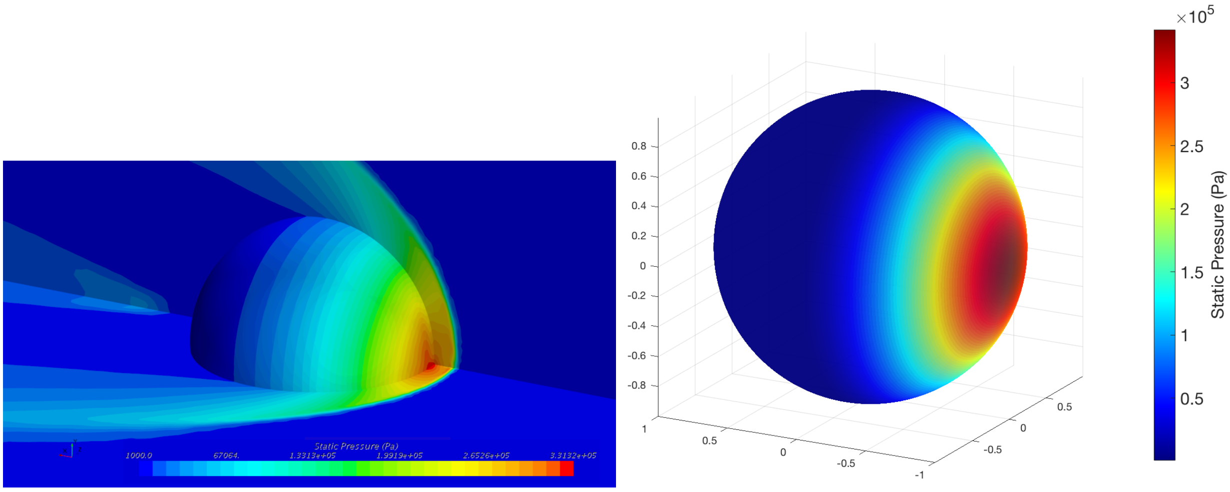
Where $\frac{P_\infty}{P_{02}}$ is found using the Rayleigh-Pitot formula (Equ. (100) in [25]) and two temperature relations may be found by Equations (43) and (95) respectively, also from [25].

The physical models were also considered. The region behind the body and shock are likely to be highly turbulent and on a larger scale than what was considered for landing. The pressure drag is likely several times greater than skin friction at high Mach numbers and so the turbulence near the wall is less important than before. With this mind, coupled flow along with the $k-\omega$ shear stress transport (SST) turbulence model was used. This model is highly studied for compressible flow [26]. This utilises the near solid-boundary accuracy of the $k-\omega$ model, and switches over to the $k-\epsilon$ model further away from the body. The $k-\omega$ SST does not encounter the sensitivity associated with freestream conditions that the $k-\omega$ model does. Other models available include the single-equation Spalart-Allmaras model (one equation rather than two reduces convergence time, but is less accurate for complex flows) and other variants of $k-\omega$ and $k-\epsilon$.

Since hypersonic flows are highly complex, convergence is difficult and generally slow to achieve. This can be managed through the use of initialisers and convergence accelerators. The solution was initialised using grid sequencing, which initialises the solution using an inviscid solver[27]. As shown in [28], an inviscid solution to a hypersonic flow field has good agreement with a turbulent one (as viscosity affects the high temperature boundary layer, rather than the whole hypersonic regime) and would be a good choice for initialisation. For coupled flow, it is suggested to use the 'expert driver' feature in STAR-CCM+ to help speed convergence of a steady simulation [29]. The Courant number was set to a low value (0.1), as per the recommendation of multiple sources, including [30].Finally, high speed flows have a very large range of velocities throughout the domain. As a result, the continuity equation is more difficult to converge than subsonic flows. A continuity convergence accelerator is included in STAR-CCM+ which helps to correct the mass flow imbalance at certain iteration intervals. It does not affect the converged solution, but does change the route reached to the solution [27]. When running the simulations however, issues with the convergence accelerator prevented further iterations after some time. This is likely due to poor calibration of the accelerator and it was decided to run the simulations for a greater number of iterations without the accelerator rather than fine-tune it.

6.3 Test Sphere

To test hypersonic flow in the CFD software, a unit sphere was tested at Mach 3 at 0° incidence and compared to the surface inclination code produced in MATLAB. A shock standoff distance for a 1 m radius spherical 'nose' was predicted to be 0.197 m by Equation 12 so the mesh was refined at this distance and behind. The computational domain was vastly reduced by considering the symmetry of the sphere. Although 2D mesh would be most efficient for an axisymmetric shape such as a sphere, the use of symmetry planes was also to be tested for the hypersonic flow (as the shuttle would be split in half by a symmetry plane). The sphere was cut horizontally and vertically into a quarter, with symmetry planes on each of the cut faces. Arbitrary freestream conditions were selected: ρ = 0.1225 kgm^{-3}, P_∞ = 1 kPa, M = 5, T = 300K. It can be seen in Figure 11 that there is excellent agreement with the two methods both numerically and visually. The stagnation point pressure P_{02} of the surface inclination code is 342 kPa and that of the CFD simulation is 331 kPa- a difference of 3.3%. Secondly, the distribution of pressure seems to be very similar. The CFD results predicts the $C_p = 0$ condition on the leeward side very well. Finally, the shock standoff distance was measured to be roughly 0.19 m, as shown in Figure 12. This agrees well with the earlier prediction of 0.20 m made using the tangent sphere method.



(a) CFD simulation

(b) Modified Newtonian code

Figure 11: Comparison of CFD and hypersonic surface inclination methods

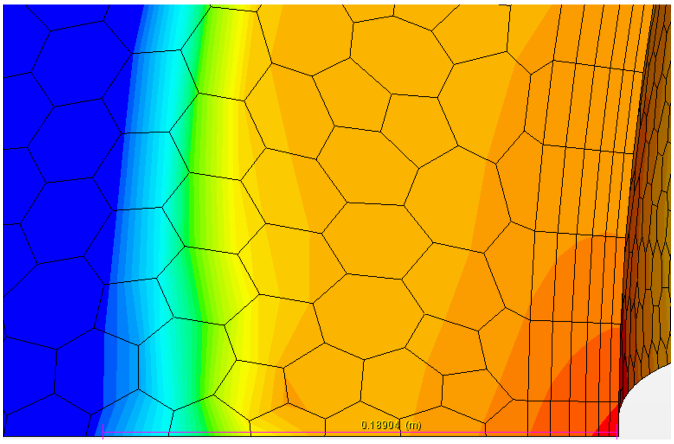


Figure 12: Shock standoff distance was predicted well

6.4 Reentry Analysis - Mach 5

After confirming that the shock behaviour of a bluff body in hypersonic flow could be captured with a suitably fine mesh, the shuttle was tested at the Mach numbers and freestream conditions in 3. Trial-and-error was used to find a realistic background mesh achievable by the desktop computers.

Figure 19 in Appendix D and Table 4 below show the results of the CFD and surface inclination code from the bottom of the shuttle. The stagnation point pressures are within 9.0% of each other. It is likely that the CFD result's pressure values are offset by the reference pressure (which was specified at the start of the solution as equal to freestream pressure), thus explaining the unreasonably low freestream pressures obtained for the simulation. With this in mind, it can be said that the CFD solution agrees well with the modified Newtonian assumption of $C_p = 0$ on the leeward side. This is shown by the dark blue region on the upper surface of the shuttle in Figure 13, indicating that the pressure here is equal to that of the freestream.

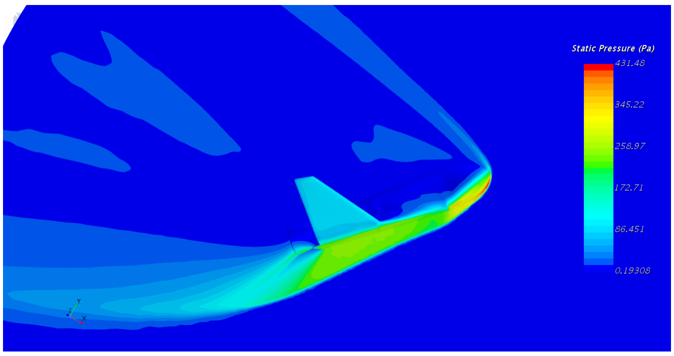


Figure 13: Pressure field showing the shock produced at mach 5

Inviscid flow solutions were also considered at the time. It is likely that the values of C_L and C_D would still remain reasonable, as shown in [28]. However, as it has been

Table 4: Agreement between the surface inclination method and the CFD simulation

	C_l	C_D
Modified Newtonian	0.3975	0.4899
CFD	0.3929	0.4805

shown, surface inclination codes provide an excellent (and far quicker) analysis of an inviscid flow. If further flow characteristics (such as the separated flow streamlines on the leeward side, as shown in Figure 14) need to be investigated, then an inviscid CFD would not provide any detail.

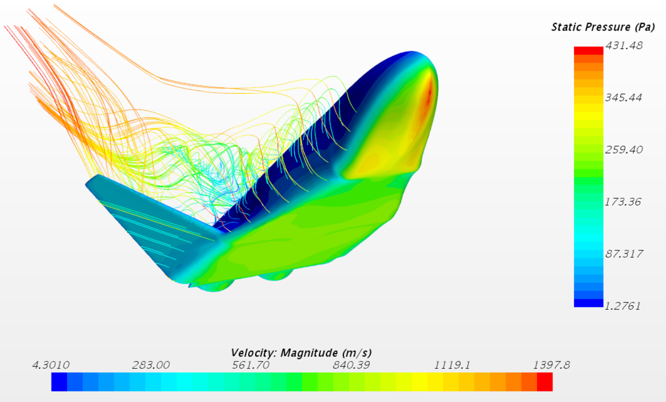


Figure 14: Streamlines showing the turbulence behind the shuttle body

The CFD simulations also ran at Mach 3, however due to time contraints, data reduction could not occur in time. Despite the efforts made in mesh refinement, it was found that the flows at Mach 7 and 10 contained pressure and velocity gradients too great to capture. Due to the mesh being too coarse for the flow, numerical errors were encountered, resulting in solutions such as that in Figure 15, where one cell has an extremely high pressure (the red prism layer cell) and almost all other cells in the domain have a negative pressure. Nonetheless, if a Mach 10 simulation at high altitude were to be run with suitable refinement, it is important to ensure that the assumptions taken at lower altitudes still hold. Most notably, the continuum assumption of the fluid is still valid. The Knudsen number, Kn defines the ratio of molecular mean free path to the length scale of the problem $(\frac{\lambda}{L})$. A low Knudsen number implies that molecules are only able to travel very short distances, meaning distances between molecules is very small and so the fluid may be assumed to be continuous. The density at the altitude when Mach 10 is achieved is very low in comparison to typical compressible flow problems. A check to ensure continuum dynamics was carried out.

The Knudsen number may alterntively be written according to Equation 13 [31].

$$Kn = \frac{M}{Re}\sqrt{\frac{\gamma\pi}{2}} \tag{13}$$

Using the values presented in 3, Re is found to be 1.68×10^6, leading to $Kn = 1.68 \times 10^{-6}$, still well within the limits of continuum dynamics. In fact, altitudes above 150 km is where the continuum assumption begins to break down. Therefore, the entire reentry analysis (which starts at 150 km) may be considered valid in this regard.

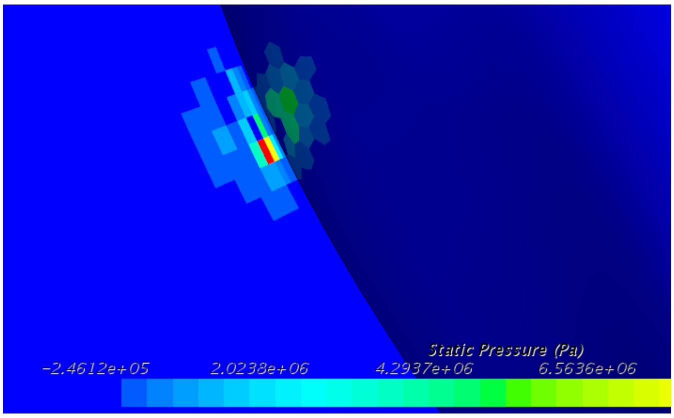


Figure 15: Insufficient mesh refinement for the Mach 10 flow.

7 Conclusions

An aircraft-launched shuttle was designed and aerodynamically analysed over the course of 5 weeks. Decisions were based on relatively simple trajectory and heating models. Now that more sophisticated models have been produced by [17] and [18], the second design iteration may use these as an input. Furthermore, another design iteration must again be evaluated with CFD, however it is hoped that the subsonic data would also be subject to verification by further wind tunnel testing. Finally, expansion beyond the modified Newtonian and tangent-wedge surface inclination methods to quickly evaluate performance at hypersonic speeds would be encouraging; especially if viscous equations are used to provide a numerical heating model. However it has been shown that these methods are already in good agreement with CFD, and far less computationally expensive.

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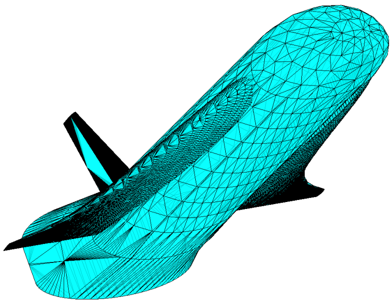
Appendices

A Surface Inclination Code Results

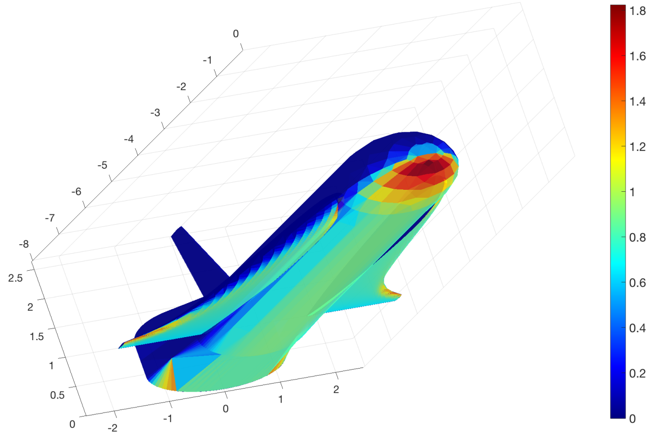
B Geometry Optimisation

C Subsonic Results

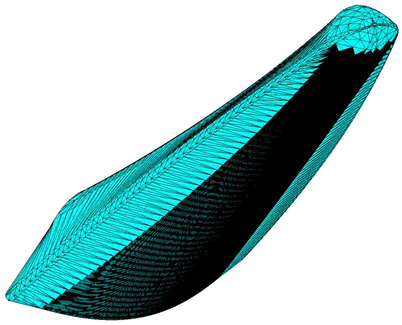
D Mach 5 Shuttle Comparison



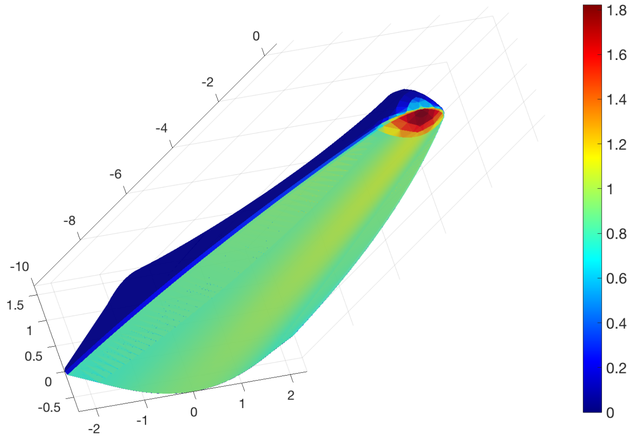
(a) Delta wing discretisation



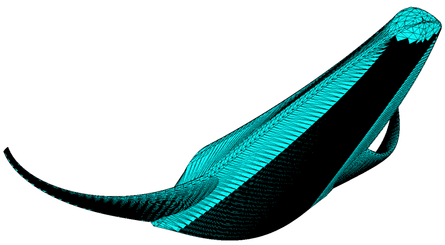
(b) Delta wing C_P distribution



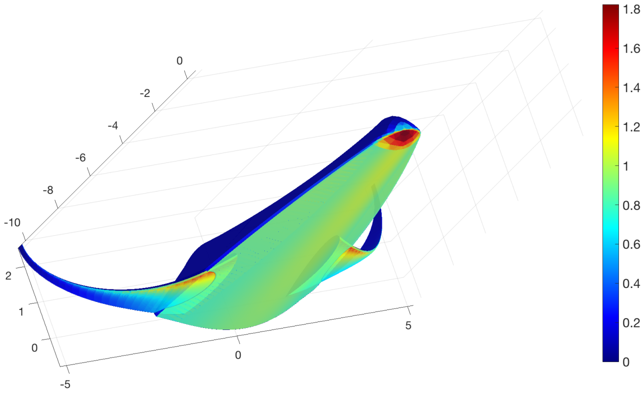
(c) HGV discretisation



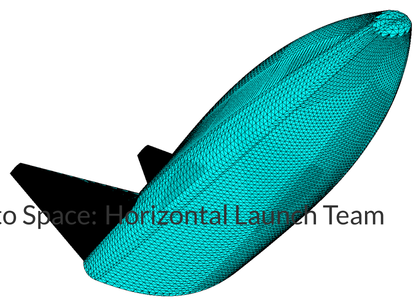
(d) HGV C_P distribution

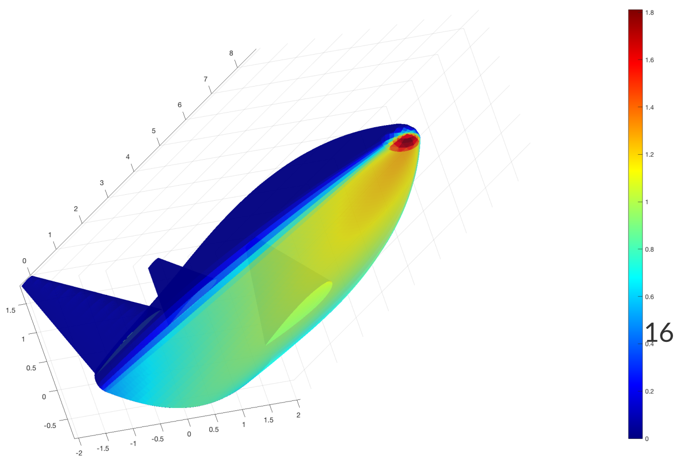


(e) Winged HGV discretisation



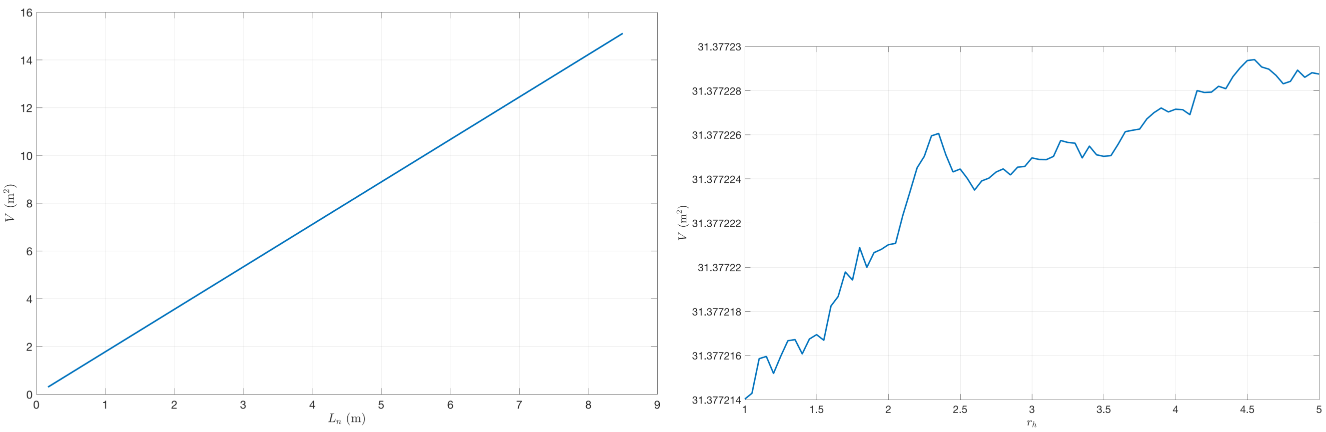
(f) Winged HGV C_P distribution





(g) BWB discretisation

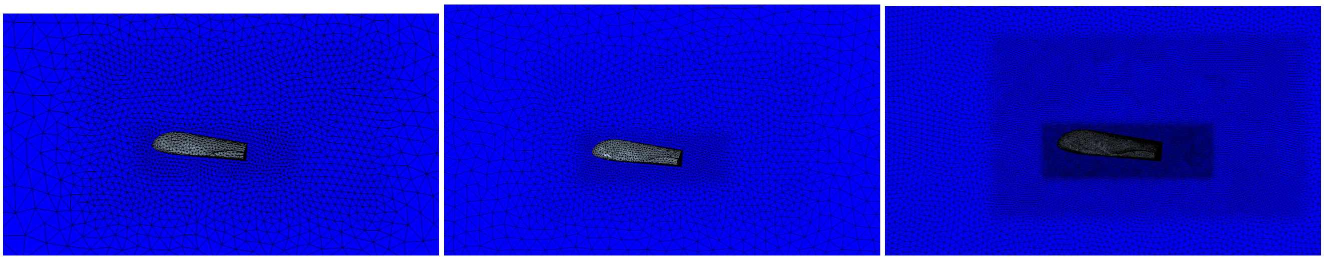
(h) BWB C_P distribution



(a) Change in nose length L_n linearly scale with volume

(b) Nose height ratio has almost no effect on volume

Figure 17: Nose volume dependence on geomtery

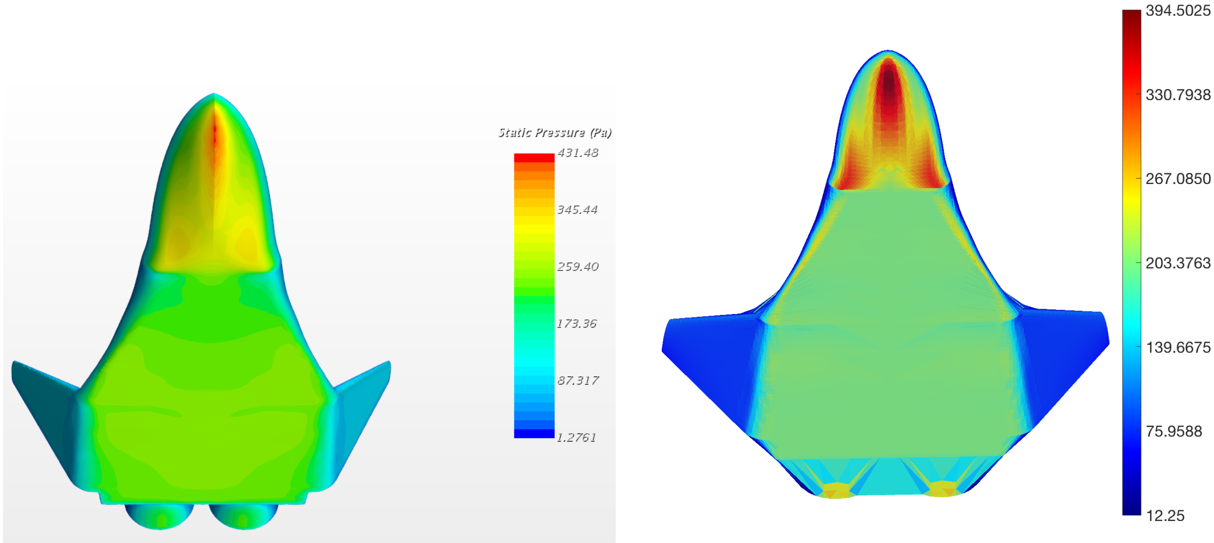


(a) Mesh 1, base size 2.0 m

(b) Mesh 5, base size 1.2 m

(c) Mesh 9, base size 0.4 m

Figure 18: Grades of mehs refinement



(a) CFD result of the pressure distribution on the shuttle

(b) Panel code result of pressure distribution on the shuttle

Figure 19: Comparison of CFD and surface inclination methods.